Logistic Regression & Nonlinear Classifiers

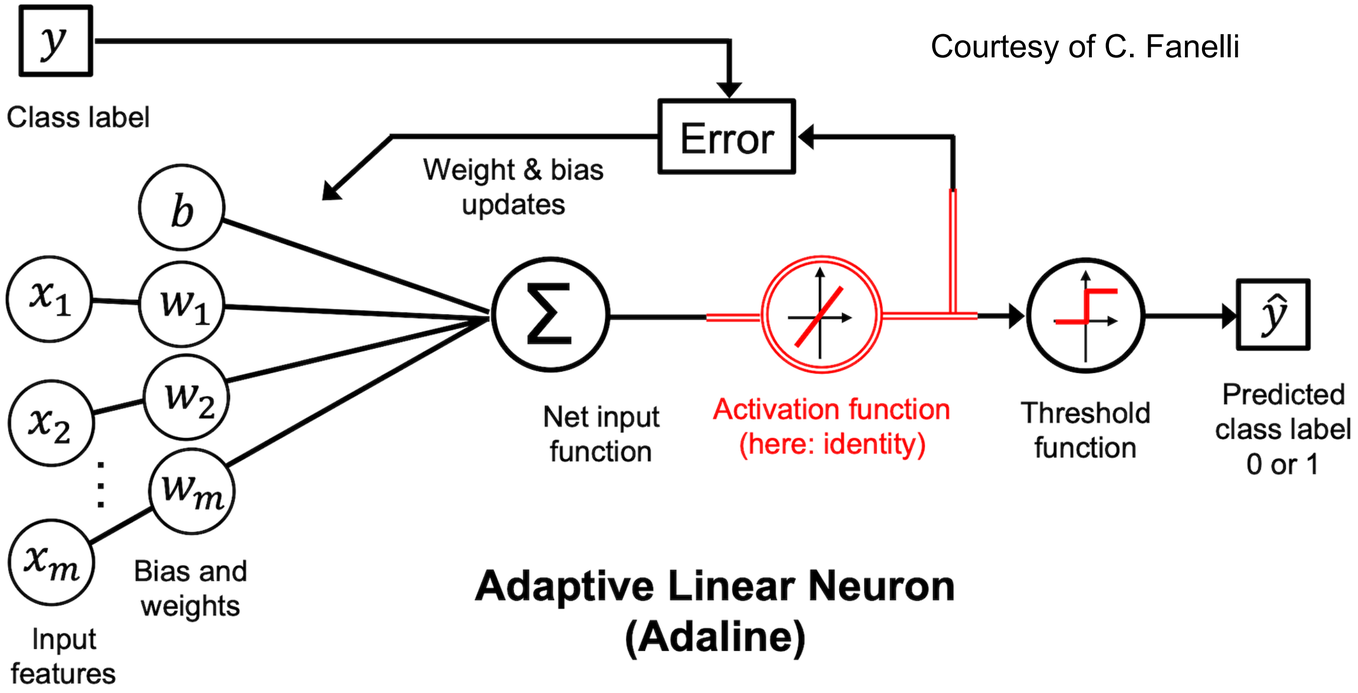
Data 442 / 610 Lecture 3

Outline

- Review of Adaline
- Gradient Descent vs. Stochastic Gradient Descent
- Logistic Regression
  - Maximum Likelihood derivation
  - Demo
  - Extensions

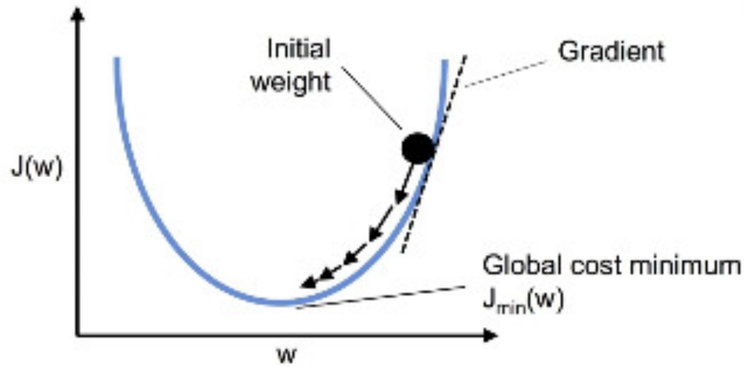
Roadmap to Deep Learning

Adaline Review



Courtesy of C. Fanelli

Adaptive Linear Neuron (Adaline)



Mean Squared Error (MSE) Loss

$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N}\left(y^i - z^i\right)^2$$

Gradient Descent: $d\mathscr{L} = \vec{\nabla}\mathscr{L} \cdot d\vec{w}$

$d\mathscr{L}$ is most *negative* when $d\vec{w}$ points *opposite* the gradient

Recall $\vec{A} \cdot \vec{B} = AB\cos\theta_{AB}$ and $\cos 180° = -1$

$$\Delta w_i = \frac{2\alpha}{N}\sum_{j=1}^{N}(y^j - z^j)\,x_i^j$$

$$\Delta b = \frac{2\alpha}{N}\sum_{j=1}^{N}(y^j - z^j)$$

Adaline: Takeaways

The good

- Update rule comes from “somewhere”
- Continuous error function allows gradient descent

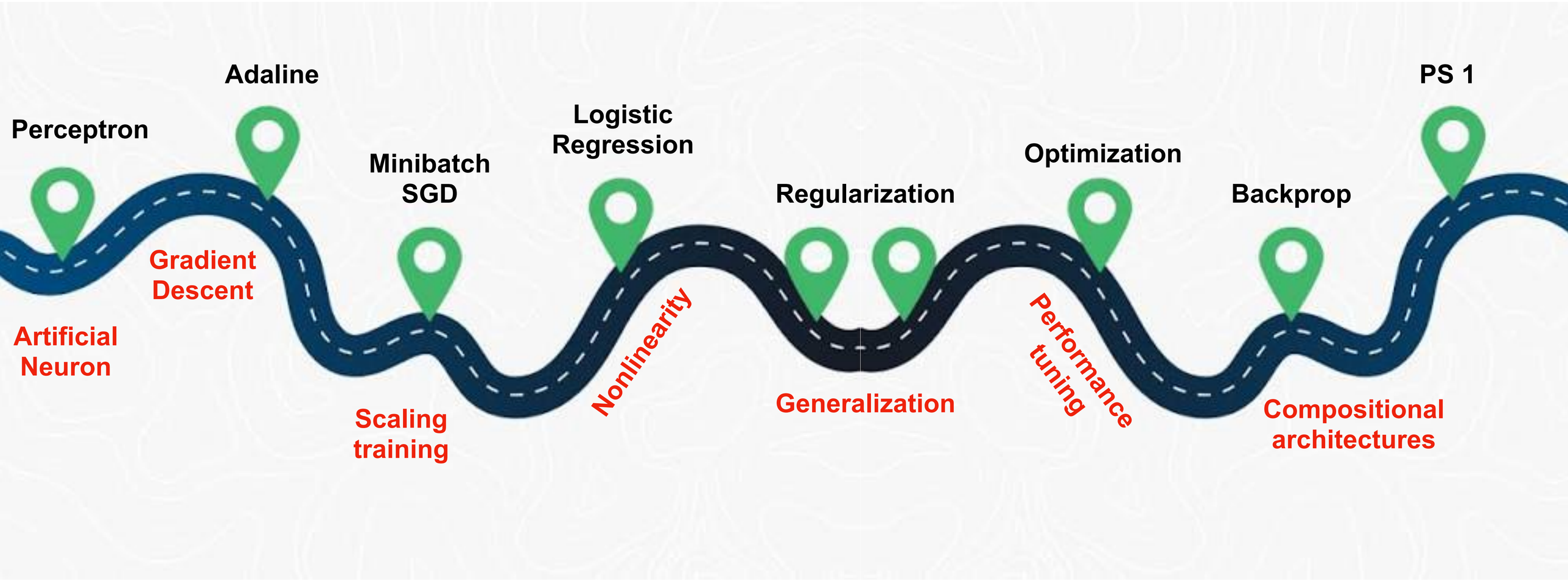
The bad

- Does not allow nonlinear decision boundaries

The ugly

- Learning rate can make or break fitting

Roadmap to Deep Learning

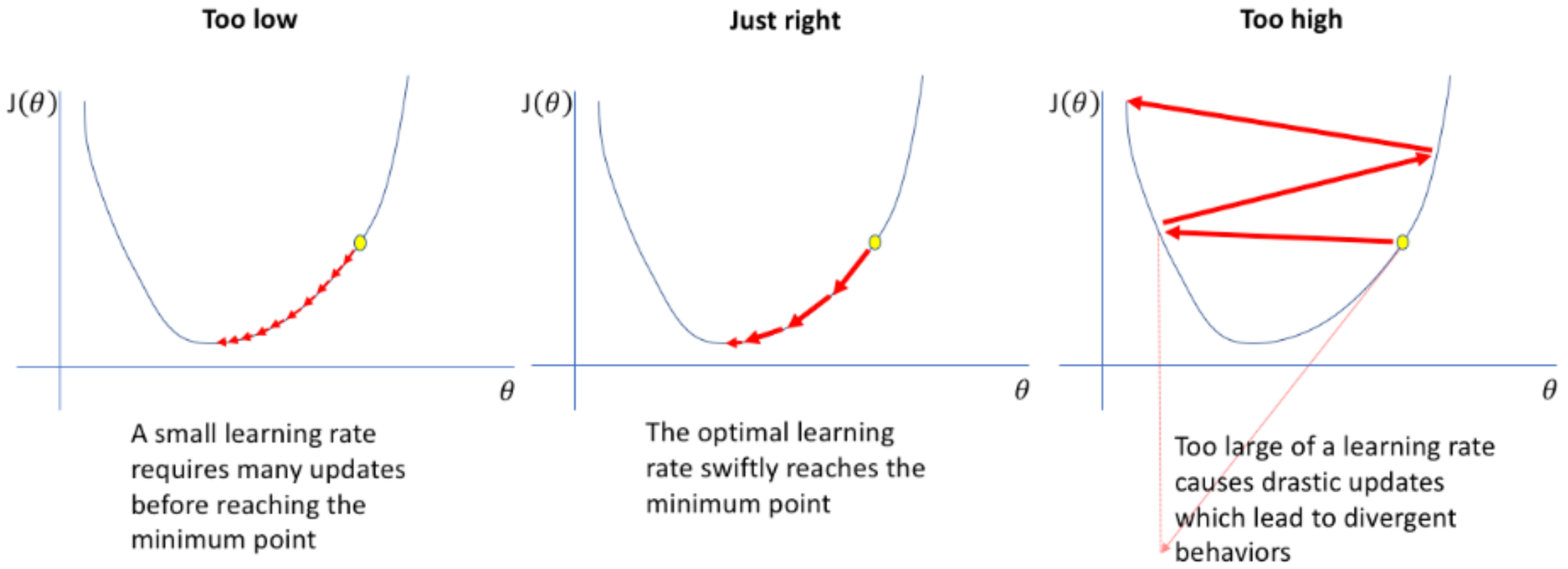


Stochastic Gradient Descent

- **Gradient descent** update rule uses the entire dataset
- **Stochastic gradient descent** updates after each data point
  - *Pros:* faster updates, stochasticity helps avoid local minima
  - *Cons:* unstable convergence, learning rate tuning

Hands-on demo: SGD training loop

The importance of learning rate



Stochastic gradient descent benefits from **learning rate decay**

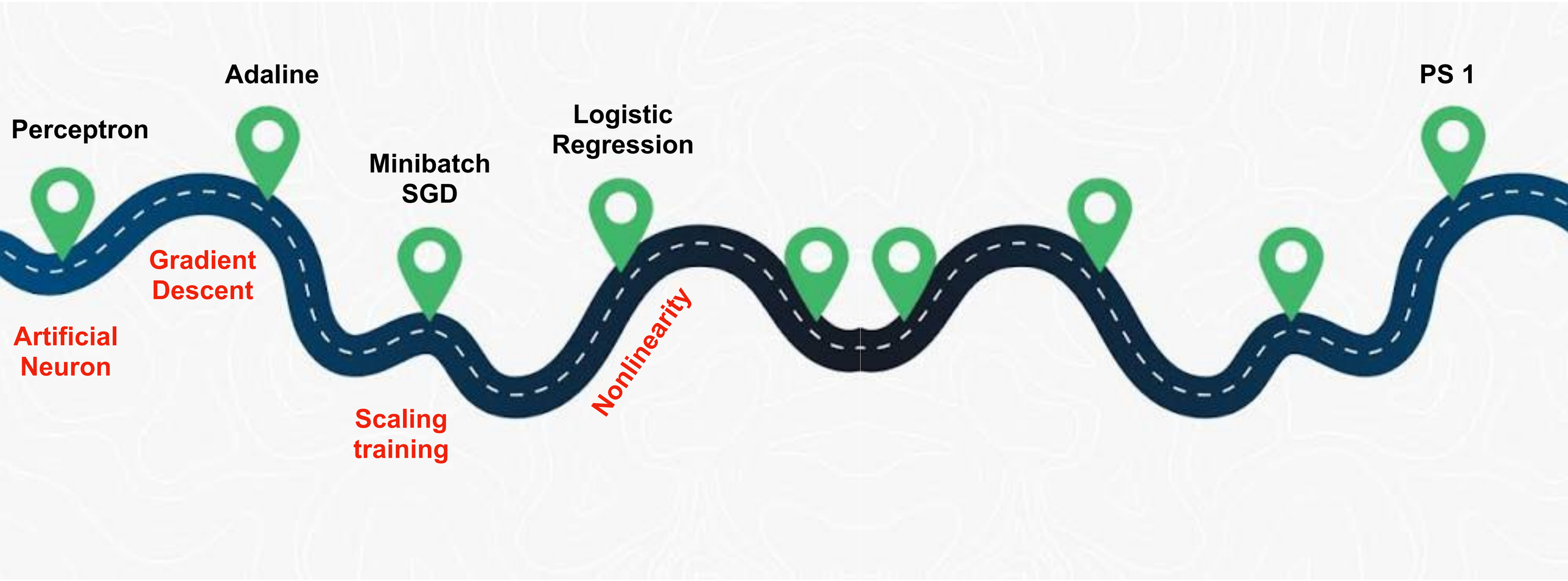
Courtesy of C. Fanelli

Stochastic Gradient Descent

- **Gradient descent** update rule uses the entire dataset
- **Stochastic gradient descent** updates after each data point
  - *Pros:* faster updates, stochasticity helps avoid local minima
  - *Cons:* unstable convergence, learning rate tuning
- **Mini-batch gradient descent** updates after each subset of data points
  - Improved gradient estimate, benefits from parallelism

Hands-on demo: Mini-batch SGD training loop

Roadmap to Deep Learning



Logistic Regression

- Adaline uses a continuous objective $\mathscr{L}$ and linear activation $\sigma(z) = z$
- Logistic regression uses a nonlinear activation to predict class probabilities
- The **preactivation** is a logit (log-odds)

$$z \equiv \text{logit}\,(p) = \log \frac{p}{1-p}$$

Preactivation: $z = \mathbf{w}^\top \mathbf{x} + b$

Sigmoid: $\sigma(z) = \dfrac{1}{1+e^{-z}}$

$$\sigma(z) : (-\infty, \infty) \to (0,1)$$

Logistic Regression: Loss function

Goal: Find $\mathbf{w}$ to maximize the *likelihood* of observations $\mathbf{w}^* = \text{argmax}_{\mathbf{w}} \left[p\left(y \,|\, \mathbf{x}; \mathbf{w}, b\right) \right]$ **Maximum Likelihood Estimation**

$$p(y \,|\, \mathbf{x}; \mathbf{w}, b) \sim \prod_i p\left(y^i \,|\, \mathbf{x}^i; \mathbf{w}, b\right) = \prod_i \sigma(z^i)^{y^i} \cdot \left(1 - \sigma(z^i)\right)^{1-y^i}$$

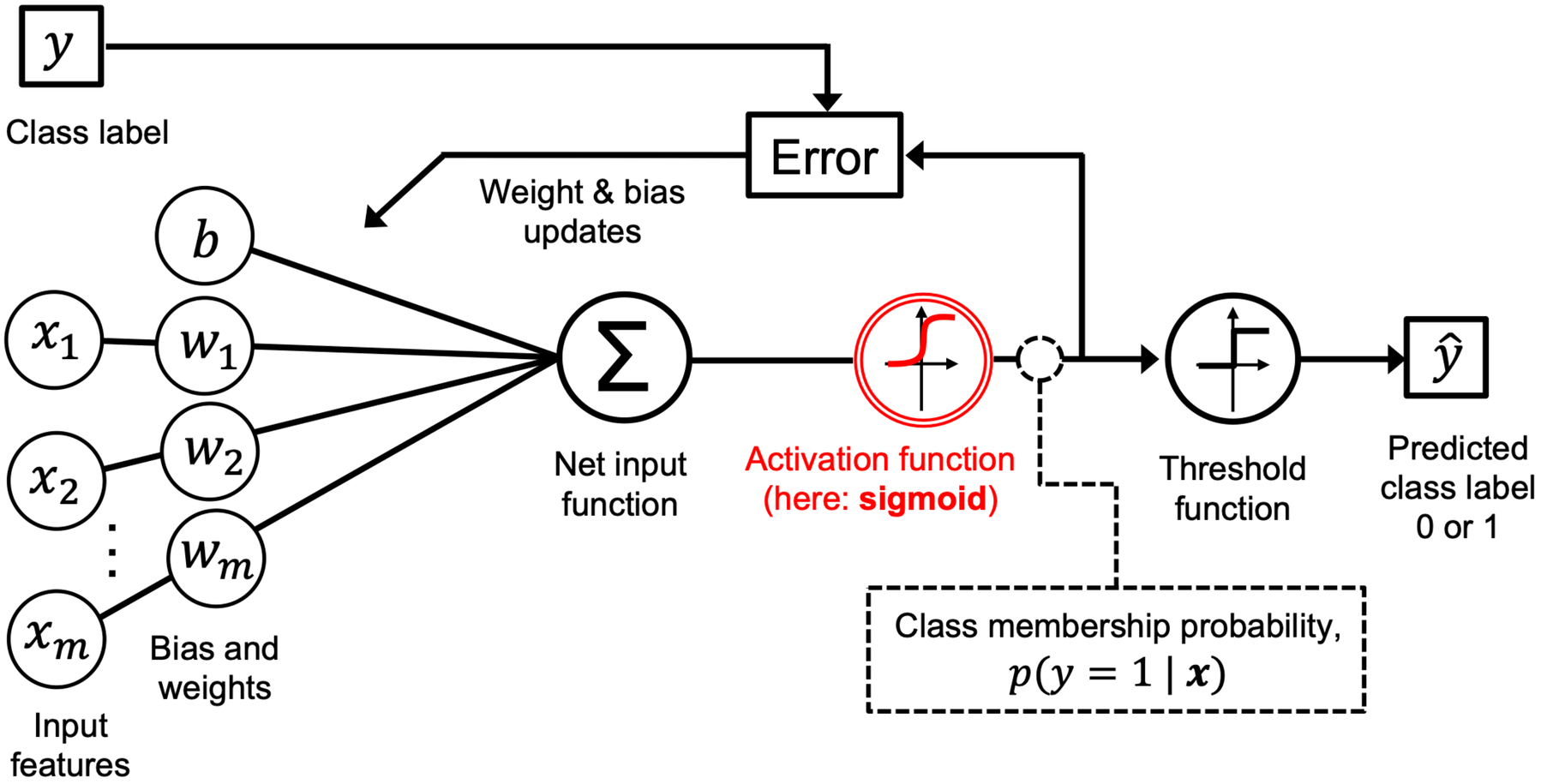
Bernoulli distribution

$$\mathscr{L} = \sum_i - y^i \log \sigma(z^i) - \left(1 - y^i\right) \log \left(1 - \sigma(z^i)\right)$$

Cross entropy loss

Goal: Find $\mathbf{w}$ to minimize the *cross entropy* loss $\mathscr{L} = \sum_i - y^i \log \sigma(z^i) - \left(1 - y^i\right) \log \left(1 - \sigma(z^i)\right)$

Logistic Regression



Courtesy of C. Fanelli

Hands-on Demo: Logistic Regression

Multiclass Regression: Softmax

- Logistic regression extends to multiple classes

$$z_j = \sum_k w_{jk} \cdot x_k + b_k$$

- Each preactivation is an un-normalized log-probability

$$z_j \equiv \log \tilde{p}$$

- The corresponding activation is the **softmax** function

$$\operatorname{softmax} z_j = \frac{\exp z_j}{\sum_k \exp z_k}$$

Numerically-stable softmax

$$\operatorname{softmax}(z_j - \max \mathbf{z}) = \operatorname{softmax} z_j$$

Views z_j as a log-odds with respect to the most-probable outcome

$$z_j \sim \log \frac{p_j}{p_{\max}}$$

That’s all!

- Slides will be posted on the course website: https://colen-teaching.github.io/DATA442-Fall-2026
- Lab demos will also be posted
- Be on the lookout for PS 0: A ***Bonus*** Assignment
- Have a great Labor Day Weekend!